


AI · MOBILITY · 2ND BATTERY · CLIMATE TECH

# Battery Trust Infrastructure for The Global Circular EV Economy

Batterfly

We help Global maximize battery **safety**, **value**, **and reuse** through AI-powered battery lifecycle intelligence.

THE BATTERY TRUST GAP

# Global lacks a trusted battery lifecycle infrastructure, Nobody can prove how healthy a battery really is

01 **Battery health is opaque & fragmented**

SoC ≠ SoH. Each OEM & swap-network sits on its own data.
No interoperable standard exists in Vietnam or ASEAN.

02 **Used-EV buyers pay a "battery uncertainty discount"**

Insurers, banks, and consumers price EV risk on vehicle-level proxies.
No battery-level data layer — no basis for trust.

03 **85% of reusable capacity is scrapped by default**

Without a grading standard, healthy second-life packs go straight to recycling the circular economy stalls before it starts.



COST OF NO TRUST

**A used EV with a healthy battery and one with a failing battery trade at the same price today.**

Distorting residual value, insurance, finance — and stalling the EU circular economy.

THREE LINES CROSSING IN 2027

# Millions of batteries will need Trusted lifecycle data by law

GLOBAL EV EXPANSION

## 17.5 M

### New BEVs Sold Globally in 2025

The global EV ecosystem has entered the era of battery lifecycle intelligence. Trust, transparency, and data-driven battery management are becoming foundational infrastructure for the **circular battery economy**.

SOURCE · ACEA 2025/2026

REGULATION

## 2027

### EU Digital Battery Passport becomes mandatory

Under Regulation (EU) 2023/1542, every EV and industrial battery > 2 kWh placed on the EU market must carry a verified digital passport from **18 Feb 2027**.

SOURCE · EU REGULATION 2023/1542

SECOND LIFE

## 12×

### Growth in second-life battery supply by 2030

Second-life EV battery supply rises from ~25–30 GWh in 2025 to ~330–350 GWh by 2030 (~65% CAGR). The market is shifting from "nice-to-have" to **infrastructure**.

SOURCE · MARKETSANDMARKETS 2025

BUILT ON PHYSICS · TRAINED ON REALITY

# One model, 17 degradation factors, Every stakeholder, One source of truth

LTE-OBD LINK · BATTERY INTELLIGENCE EDGE DEVICE




01 · INPUT

## Real-driving CAN data

Years of CAN-level data from production fleets — not lab cycles. OBD telemetry, charge/discharge, thermal and duty-cycle signals.



02 · ENGINE

## 17-factor weighted ML Model

Patent-backed model combining electrochemistry priors with continuous-learning ML. Multi-factor degradation, thermal-risk prediction, RUL estimation.



03 · OUTPUT

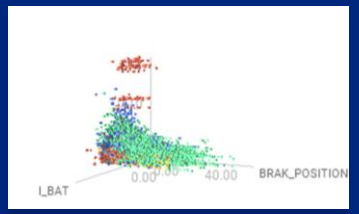
## SOH · RUL · RV Certificates

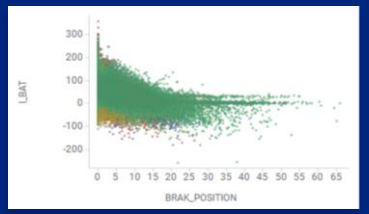
Battery Health Certificates, fleet dashboards, residual-value & RUL reports, insurer/lender APIs, Battery Passport schema.

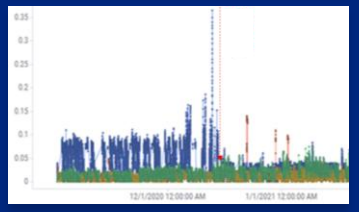


DEFENSIBILITY · ACCURACY OVER TIME

## A data moat that compounds.



Every vehicle-year of operating data sharpens the model : a self-reinforcing loop no incumbent can copy by writing software.

SEVEN YEARS OF DATA · FOUR FLEET DOMAINS · ONE ALGORITHM

# From a single pack to a national fleet, Our algorithm grew with the data

**OUTCOME**

**EV battery lifecycle data collected, analyzed, and productized into a deployable algorithm**

**2026**

### Metro Rail Battery Safety Monitoring

Seoul Metro · rail-grade battery operation telemetry & control

**2023–2025**

### Electric Taxi Driving & Charging Big Data

2-yr dataset · drive + charge pattern fusion · **validation complete**

**2021–2022**

### Electric Bus Route Big Data

**17 degradation factors** isolated · prediction algorithm

**2019–2020**

### EV Battery Lifecycle Management System

Blockchain-anchored lifecycle ledger · stage-by-stage data capture

DATA SCALE · COMPOUNDING ASSET

## Every fleet domain adds a new training distribution.

| CAN SIGNALS | FLEET DOMAINS |
|---|---|
| 17 | 4 |
| **SOH ACCURACY** | **LIFECYCLE STAGES** |
| **98.26%** | 3 |

**NEXT — EU EXPANSION**

Same playbook, EU duty cycles: **Luxembourg pilot → fleet partners → Battery Passport schema integration.**

THE BATTERFLY PLATFORM

# One platform covering the Entire battery lifecycle

No battery removal. No lab cycles. **Real-driving CAN data → AI diagnosis → tradable certificates → second-life routing**

MODULE 01 · **WEDGE**

## DEAL4EV

**AI Battery Diagnosis & Certification**

MODULE 02 · **PASSPORT**

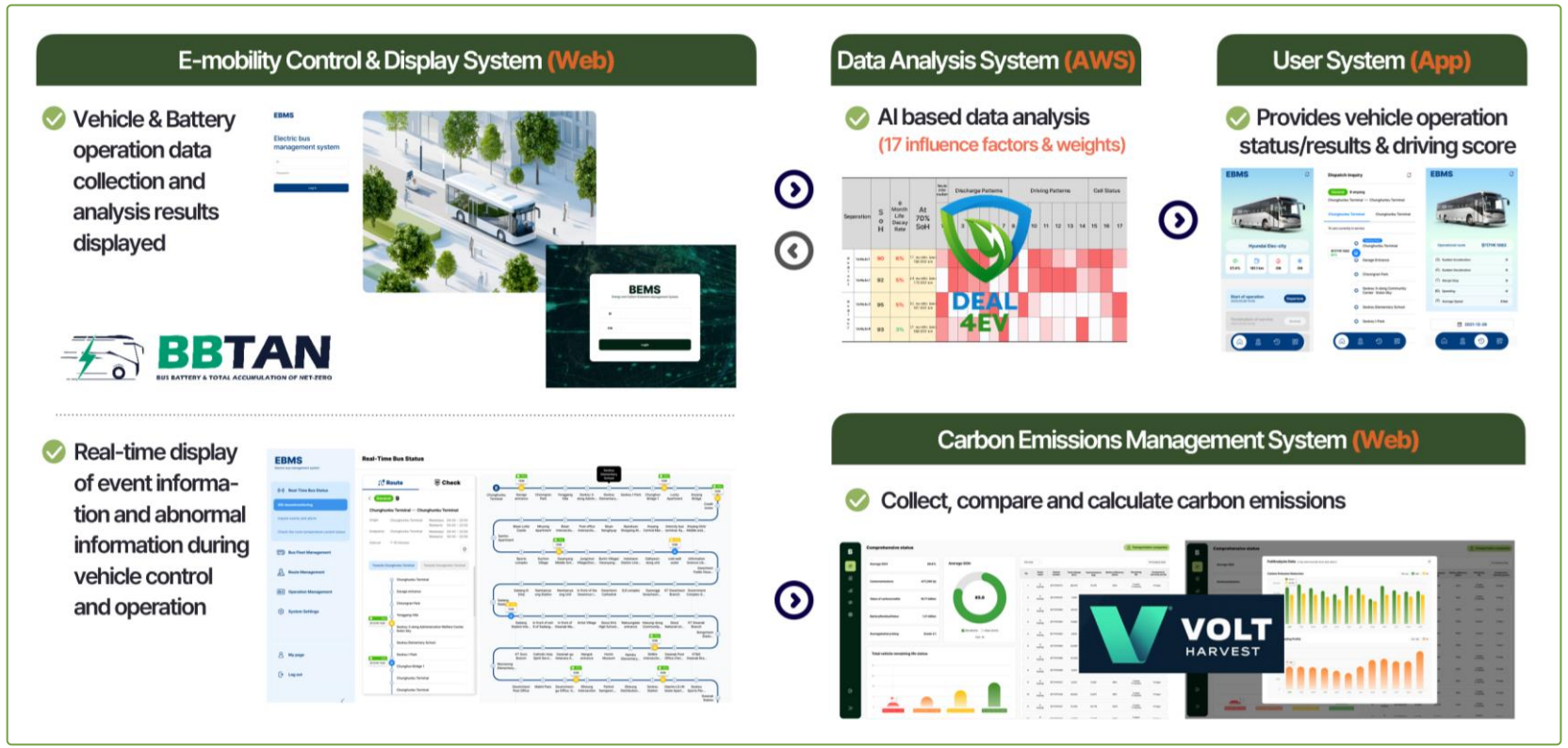
## BBTAN

Fleet Management & Traceability

MODULE 03 · **CIRCULAR**

## VOLT HARVEST

Second-Life Battery Marketplace



HOW IT WORKS : DEAL4EV

# Continuous, structured diagnostics
# via Battery Intelligence Edge Device



## LTE-OBD Link for DEAL4EV

Battery Intelligence Edge Device

**98.26%** acc. **300+** shops **24h** turn

**01** **OBD-II install**
Install OBD-II and verify the link.
✓ ~5 minutes

**02** **Real-time CAN ingestion**
Stream driving and charging data.
✓ Real-time

**03** **AI analysis & prediction**
Anomaly · SOH · life prediction
✓ 98.26% accuracy

**04** **Report & shop link**
Report + shop link.
✓ 100+ shops

**05** **Certificate issued**
KFMSA-registered certificate.
✓ Official

**Process time** ≤ 24h total

**Accuracy** 98.26%

**Partner shops** 100+

**Warranty** 6-month cover

A systematic 5-step program builds battery diagnostic certifications.

SIX STACKED REVENUE STREAMS

# One battery. Six recurring revenue streams. A scalable platform for the entire battery lifecycle.

TRANSACTIONAL · LIVE

## Battery Diagnostics

**$60** per scan

TRUST LAYER

## Certification Services

**$80** finance-grade cert

RECURRING · ARR

## SaaS Subscription

**$12** / vehicle / month

REGULATION · 2027

## Battery Passport

**$30** + $6 / yr hosting

FINANCE · API

## Residual Value API

**$60k~$600k** / partner / yr

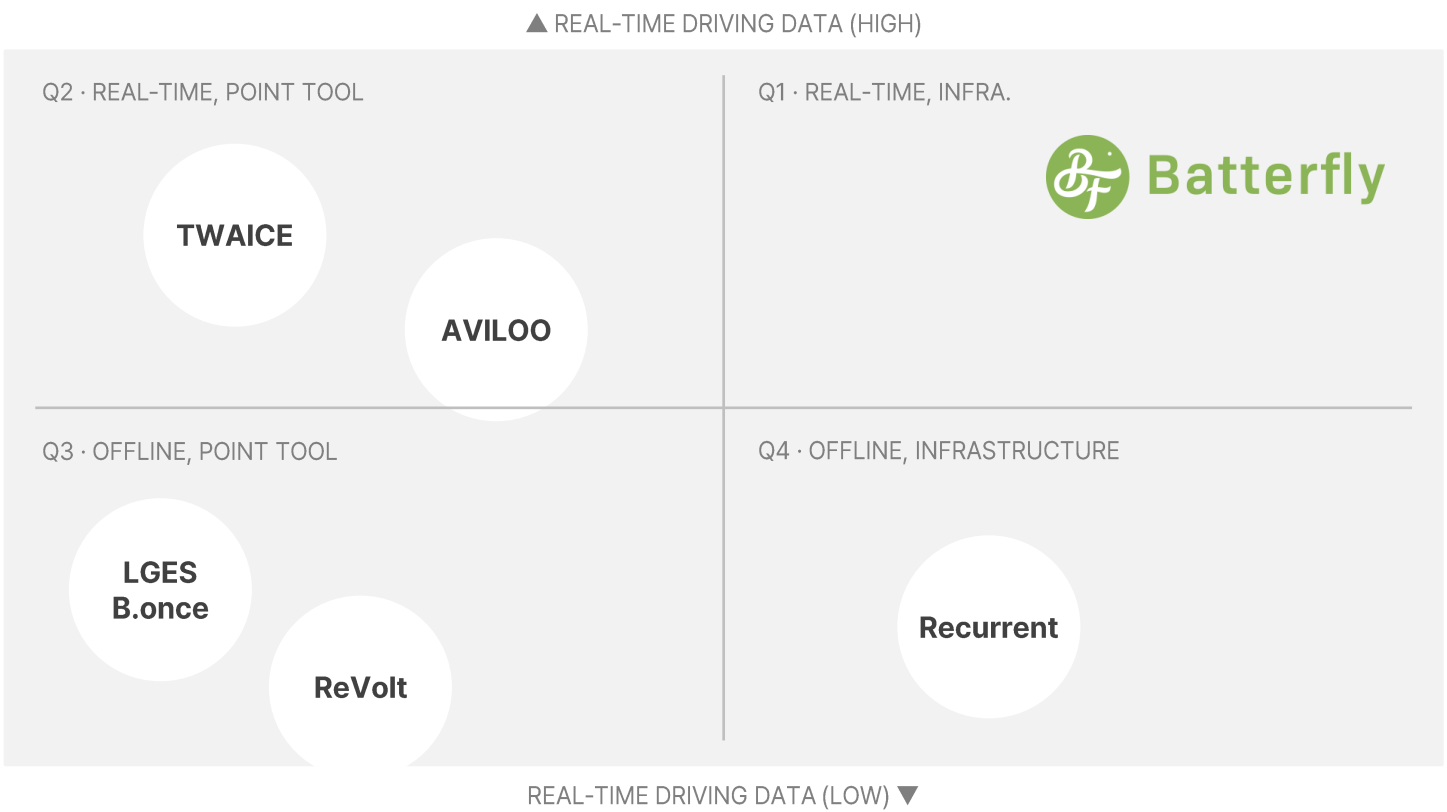
CIRCULAR · 2ND-LIFE

## Marketplace Commission

**3~5%** of transaction value

COMPETITION

# Competitors analyze batteries,
# BatterFly builds battery infrastructure




## CORE DIFFERENTIATORS

### Lifecycle Data Layer

Diagnostic → authentication → transactions → insurance → reuse integrated data — competitors focus on just one point

### Repair-Network Distribution

Directly owns the repair shop channel — TWAICE· AVILOO relies on OEMs and dealers

### Local Driving Data

Korean Road and Climate Learning Data — Recurrent· No global operators like AVILOO

### Official Certification Standard

KFMSA Co-Issuance — A Standard Referenced by Transactions, Finance, and Insurance

**+ Accuracy 98.26% · DATA Class A**

Data MOAT

# Battery data Over time, It becomes a barrier to entry

Using Korea's diverse climate and driving conditions as a testbed,
**we are creating Global-ready battery intelligence powered by real-world EV battery data**

| MOAT 01 | MOAT 02 | MOAT 03 | MOAT 04 | MOAT 05 |
|---|---|---|---|---|
| **Real Driving Data** | **Regional Battery Behavior** | **Cross Lifecycle Dataset** | **Repair Network Distribution** | **Continuous Learning AI** |
| Actual driving data reflecting Korean roads, climate, and driving patterns as is, not laboratory data | Degradation patterns by region, season, and charging infrastructure — user-side datasets OEMs cannot own | A single dataset connected throughout the entire lifecycle, from driving → diagnostics → maintenance → transactions → reuse | Data distribution through workshop channels — a structure that simultaneously occupies transaction and financial touchpoints | A self-reinforcing AI learning loop that increases accuracy as data accumulates |
| ACCUMULATED<br>7 YEARS+ | REGIONS<br>6 KR | STAGES LINKED<br>5 / 5 | PIPELINE<br>2,800 GOAL | CURRENT ACC.<br>98.26% |

UNFAIR ADVANTAGE & TEAM

# Built on decades of expertise.
# Trained on millions of real-world battery data points.
# Trusted by leading industry partners.



## FOUNDER & CEO

**Sungwon Park**

30+ yrs. ICT · 10+ yrs battery industry.
Built EV Battery Life Cycle business.
Led national EV operating-data programs.

**Spin-Off** from LG



## COO

F.I.D.M (USA)
National R&D PM
Bachelor of Eng.



## CTO

LG & Startup (31yr.)
SI & Implementation Specialist



## CMO



## CSO



## CDO

## Companies & institutions of MOU

LG CNS · LG U+ · Seoul Metro
EASY CHARGER · 타고 tago.kr · 삼영운수주식회사
Korea Transportation Safety Authority · 전라남도 JeollaNamdo · 보영운수주식회사
사단법인 미래항공모빌리티산업협회 · 환경부 Ministry of Environment · 국토교통부 Ministry of Land, Infrastructure and Transport
KFMSA Korea Future Mobility Service Association · 산업통상자원부 Ministry of Trade, Industry and Energy · 과학기술정보통신부 Ministry of Science and ICT

SUNG KYUN KWAN UNIVERSITY · Yeungnam University · CHONNAM NATIONAL UNIVERSITY




# We are the Battery Doctor !

Building AI & Bigdata that deliver

**accurate battery health diagnostics** for every

**EV** / **Driver** / **Climate** / **Road** in the world.

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# Market Opportunity

Battery lifecycle value unlock = massive market potential across resale, reuse, and recycling

## Global Market Size

Battery reuse & second-life market

| $20B | 52.3% | 80M | $15B |
|---|---|---|---|
| Market by 2030 | Used EV CAGR | Units by 2030 | Fleet Market |
| +15% CAGR | 2024-2030 | Global scale | B2B segment |

**Batterfly is the trust infrastructure across resale → reuse → recycling**

Our platform enables battery lifecycle value creation through AI-powered diagnostics and blockchain verification

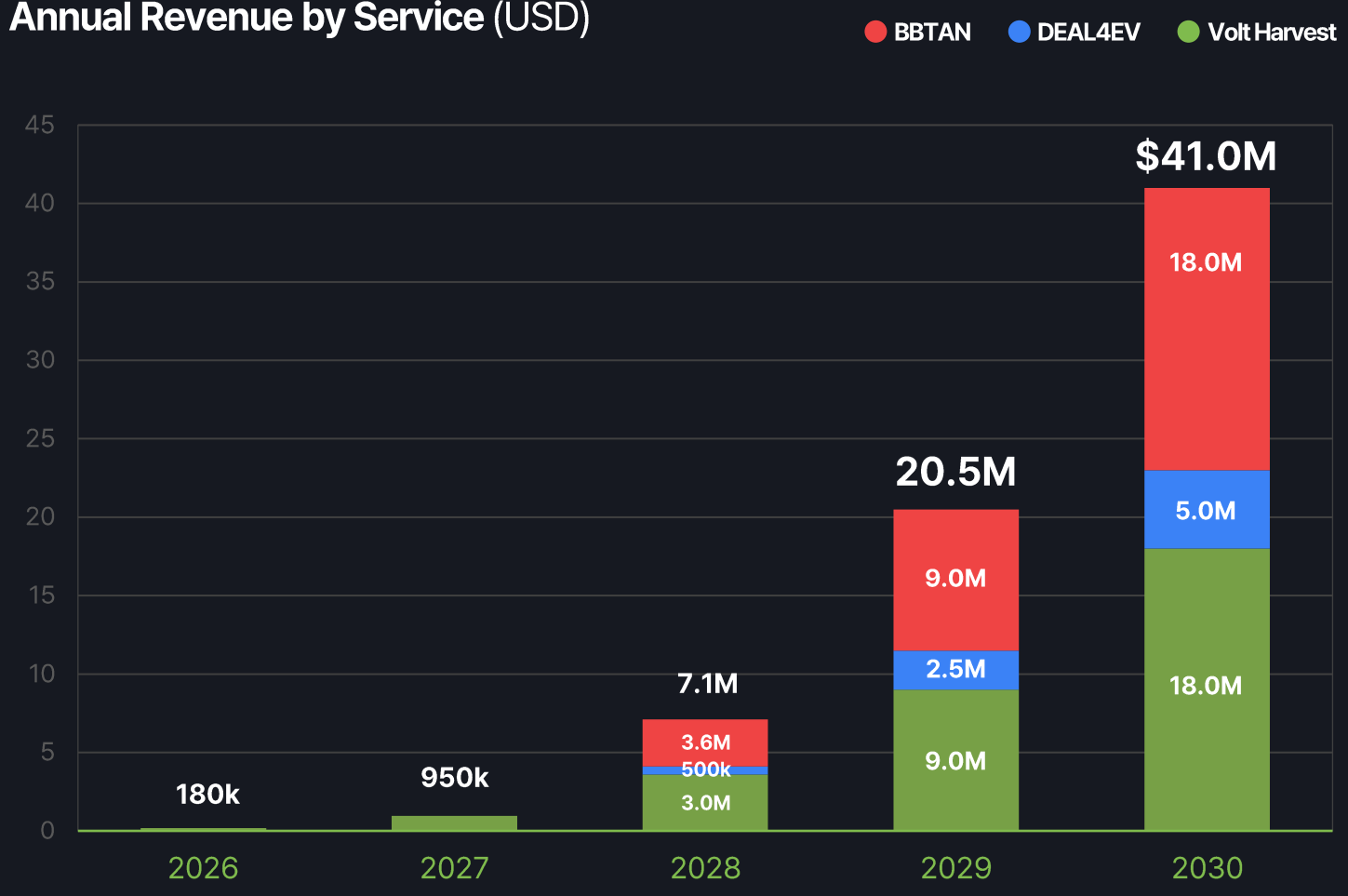
## Market Segments

Revenue breakdown by category

| | |
|---|---|
| Used EV Sales | $8.2B (41%) |
| Battery Reuse | $6.5B (32%) |
| Recycling | $3.3B (16%) |
| Maintenance | $2.0B (10%) |

# Revenue Projection 2026-2030

## Annual Revenue by Service (USD)




5-Year Cumulative Revenue

**$70M**

2026 - 2030 Total

## Growth Drivers

**2028 Tipping Point** — Key Milestone

Revenue jumps 7.5x with the launch of Volt Harvest ($3k/unit) entering the market.

**Stable Recurring**

BBTAN provides consistent baseload revenue growing to $18M by 2030.

**Mass Adoption**

DEA4EV scales to 100k+ uses, capturing high-volume diagnostic market.

2030 Target **$41.0M** — Exponential

# Who is Customer

Trusted Battery Intelligence Is Essential Across the EV Ecosystem. (Revenue growth to $ 460K in ‘25.)

## B2B Market (PoC)

**Electric Vehicle Manufacturer**
(EV Bus of Hyundai, Edison Motors)






**Electric Bus Route Transport Company**
(Samyoung, Boyoung)

**Robot of Smart Fulfillment Center**
(Logistics Center, LG CNS)

**OEM Battery Maker, Fleet Operator, Service Center & Dealer, Used EV Market, Insurance**

## B2G Market (Project)

**EV Battery Data BM Demonstration**
(Local Goverment, Jeollanam-do)

**Carbon-neutral Integrated Environmental Information Center Construction** (MOE)

**Development of Advanced Technology for Carbon-neutral Transportation Sector Reduction Strategy** (MOLIT)

**Sustainable Regional Energy Ecosystem Project in** (Busan Smart City)

**AI-Based Real-Time Management System for Railway Vehicle Batteries** (Seoul Metro)